



Per- and polyfluoroalkyl substances (PFAS) in final treated solids (Biosolids) from 190 Michigan wastewater treatment plants

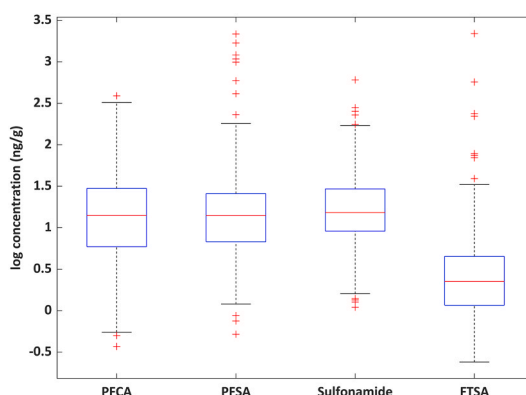
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HIGHLIGHTS

- Analysis of 350 biosolids from geographically-diverse set of 190 WWTPs.
- Primary sample variability attributed to long chain PFAS.
- Secondary sample variability attributed to short chain carboxylic compounds.
- Collective concentrations of sulfonamide precursors often greater than PFOS.
- Trends in PFAS enrichment generally increase with compound hydrophobicity.

GRAPHICAL ABSTRACT



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ABSTRACT

Trends in concentration, distribution, and variability of per- and polyfluoroalkyl substances (PFAS) in biosolids are characterized using an extensive dataset of 350 samples from 190 wastewater treatment plants (WWTPs) across Michigan. All samples are comprised of final treated solids generated at the end of the wastewater treatment process. Concentrations of both individual and Σ_{24} PFAS are lognormally distributed, with Σ_{24} PFAS concentrations ranging from 1–3200 ng/g and averaging 108 ± 277 ng/g dry wt. PFAS with carboxyl and sulfonic functional groups comprise 29% and 71% of Σ_{24} PFAS concentrations, respectively, on average. Primary sample variability in concentration is associated with long-chain PFAS with higher tendency for partitioning to biosolids. Short-chain carboxylic compounds, most notably PFHxA, are responsible for secondary concentration variability. Usage of FTSA and PFBS replacements to long-chain sulfonic compounds also contributes to variance in biosolids concentrations. Sulfonamide precursor compounds as a collective group are detected at a similar frequency as PFOS and often have higher concentrations. Trends in PFAS enrichment for individual PFAS vary at least 3 orders-of-magnitude and generally increase with compound hydrophobicity; however, partitioning of PFAS onto solids in WWTPs is a complex process not easily described nor constrained using experimentally-derived partitioning coefficients.

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1. Introduction

Per- and polyfluoroalkyl substances (PFAS) are a diverse class of highly persistent, organic environmental pollutants containing at least one carbon-fluorine bond and cause adverse human health impacts at parts-per-trillion concentrations. Polyfluorinated precursor PFAS, most of which are excluded from analytical suites used for regulatory compliance, can transform to terminal perfluoroalkyl acids (PFAAs) in the environment [1-4], which are detected. PFAS, particularly terminal PFAAs, persist in the environment because the high strength of the carbon-fluorine bond resists natural degradation mechanisms [5-7]. As a result, PFAS cycle between anthropogenic treatment and disposal activities, such as wastewater treatment plants (WWTPs) and landfills, and the natural environment [8-13]. Of the thousands of species presently identified, Buck et al. [14] identified 256 commercially relevant PFAS compounds. Only two PFAS, both PFAAs, are widely regulated and monitored in drinking water and wastewater; perfluorooctanoic acid (PFOA) and perfluorooctane sulfonic acid (PFOS) [15-17]. However, dozens of other PFAS compounds are detected worldwide in every environmental medium [18-22]. In response, PFAS release mechanisms and pathways to the environment are receiving increased attention [12, 23,24].

WWTPs receive PFAS from myriad sources and treated effluent and solid residuals are an important pathway of PFAS release to the environment [9,25-28]. Sources of PFAS in wastewater influent include consumer products, landfill leachate, aqueous film-forming foam (AFFF) from municipal, commercial, and military activities, and industrial waste streams [4,11,29-32]. Many precursors are readily transformed to PFAAs in WWTPs [9,33], but not all precursors are fully transformed to PFAAs, and the PFAAs are not degraded in WWTPs. Due to their tendency to partition to the solid organic phase, multiple PFAS accumulate in the final treated solids generated at WWTPs [25,27,34-39]. Partitioning of PFAS to the solid organic phase has been shown to increase with increasing carbon chain length due to increasing hydrophobicity [34,40].

Disposal of the large quantities of final treated solids (Fig. S1) generated in wastewater treatment is one of the most expensive problems facing WWTPs. The least expensive and most sustainable option is the beneficial reuse of final treated solids (biosolids¹) as a soil amendment in agriculture. In 2019, roughly 2.44 million dry metric tons (51%) of biosolids generated in the U.S. were land applied [17]. However, elevated concentrations of PFAS have recently led to regulations aimed at reducing PFAS loading to agricultural soils [41-43]. In 2022, the U.S. State of Maine introduced a complete ban on the land application of biosolids due to concerns about PFAS [44].

The application of PFAS-laden biosolids can potentially contaminate soils, crops, livestock, dairy products and shallow groundwater systems [19,26,45]. Several studies have noted the ability of plants including agricultural crops to uptake bioavailable PFAS (particularly shorter-chain compounds) within soil [19,46-49]. Once concentrated in plants, the consumption of plants laden with PFAS by livestock can also lead to further bioaccumulation in animal tissue and dairy products, further increasing the complexity and diversity of PFAS exposure pathways within the food supply [49-51]. Land application of PFAS-contaminated biosolids could also potentially lead to the contamination of water resources as PFAS become mobilized through biosolids degradation and leaching [26,52-55].

Biosolids that are not land applied are disposed of in landfills or incinerated, both of which are more expensive options. Similar to WWTPs, landfill environments enable the transformation of polyfluorinated precursors to PFAAs, but does not break down PFAAs [56,

57]. As a result, landfill disposal of biosolids likely contributes to the compositions and concentrations of PFAS in landfill leachates [9,10]. The offsite treatment of landfill leachate at WWTPs establishes a perpetual PFAS cycle between landfills and WWTPs that increases the potential for PFAS discharges to the environment. Emerging treatment technologies for PFAS-contaminated biosolids may prove useful in disrupting this cycle [58,59].

Several studies have investigated the distribution and composition of PFAS in biosolids-amended soils [26,28,42,60,61]. However, few have characterized PFAS occurrence and distribution in biosolids prior to land application. Because every land-application site is unique in its soil composition and climate, it is laborious and often cost-prohibitive to compare and evaluate PFAS signatures of land-applied biosolids for a multitude of sites. Analyzing PFAS signatures of freshly generated samples of biosolids at the end of the wastewater treatment process is a more practical approach because the results can be universally applied to different soils and climatic conditions and to other disposal methods, such as landfilling and incineration.

Most previous investigations in the literature have concluded that long-chain sulfonate PFAS compounds, such as PFOS, tend to be the dominant analytes detected in biosolids [27,28,39,62]. However, some short-chain carboxylate PFAS, such as perfluorohexanoic acid (PFHxA), are also highly detected [30,63]. Higher detection of PFAAs is partially due to microbial-facilitated transformation of precursor compounds such as perfluoroalkane sulfonamido and polyfluorinated telomer substances to PFAAs in biosolids/sludge at various stages in WWTPs [64-66]. Precursor transformations mainly occur during secondary treatment but the extent of degradation towards PFAAs varies widely seasonally and between WWTPs [64,65]. Significant transformations have also been observed during anaerobic and aerobic sludge digestion [65]. Therefore, final solids treatment and storage time are important parameters to consider when comparing PFAS partitioning behavior between WWTPs, in addition to partitioning and transformations that occur during wastewater treatment. Transformations are also expected to continue after land application or disposal in a landfill.

Higgins et al. [62] pioneered the characterization of PFAS in solid matrices by investigating the presence and concentrations of 12 PFAS compounds in sediments and biosolids/sludges. Despite the rapid progression of PFAS research, the scope of Higgins et al. [62] remains comparable to modern biosolids studies. However, along with the studies that succeed Higgins et al. [62], investigations of PFAS in biosolids have largely been limited in sample population [8,28,36,38,67, 68] and/or analyte suite [8,28,39,68,69] (Table 1 and S1). One of the largest sample populations in the literature (4981 samples from 1165 WWTPs) was analyzed by Ulrich et al. [70], but the analytical suite was limited to 11 PFAS compounds. Notably absent from the analyte list were several sulfonates, such as perfluorosulfonic acids (PFSAs), fluorotelomer sulfonates (FTSA), and perfluoroalkane sulfonamido substances, particularly N-MeFOSAA. Many researchers have demonstrated that these sulfonic compounds, which are PFOS precursors, constitute a major portion of PFAS in biosolids [27,28,62,71]. Lastly, some studies included samples from storage banks which do not reflect the PFAS compositions and signatures of modern biosolids [27].

This study investigates the concentration, distribution, and variability of PFAS in final treated solids (biosolids) from the largest known sample population collected in the U.S. to date, representing a geographically-diverse range of WWTPs serving communities of various sizes and industrial activity. Solids samples were collected from 190 Michigan WWTPs and analyzed for a suite of 24-29 analytes (depending on the year) that represent 7 PFSAs, 11 perfluorocarboxylic acids (PFCAs), 3 sulfonamide compounds (N-MeFOSAA, N-EtFOSAA, and FOSA), 3 FTSA, as well as HFPO-DA (GenX), ADONDA, 9Cl-PF3ONS, 11Cl-PF3OUdS, and PFecHS (Table S2) [89]. Multivariate and univariate statistical analyses are used to assess trends in carbon chain length and functional group, determine concentration ranges and detection rates, and evaluate compound variance. PFAS enrichment in biosolids

¹ Our definition of biosolids refers to all final treated solids generated at the end of the wastewater treatment process and differs from the terminology used by the state of Michigan as defined in Section 2.2.

relative to WWTP influent is evaluated for biosolids-influent pairs.

2. Methodology

2.1. Data procurement

The Michigan Department of Environment, Great Lakes, and Energy (EGLE) has performed several statewide sampling campaigns and investigations of WWTP influent, effluent, and final treated solids [25,90]. The first of three major investigations, the 2018 Evaluation of PFAS in Influent, Effluent, and Residuals of WWTPs in Michigan, included 85 aqueous and 71 solids samples from 42 WWTPs [25]. A second investigation, the 2021 Statewide WWTP and Biosolids/Sludge Study, included 85 aqueous (influent and effluent) and 44 solids samples from 37 WWTPs. A subset of 22 WWTPs are common to both the 2018 Evaluation of PFAS in Influent, Effluent, and Residuals of WWTPs in Michigan and 2021 WWTP Statewide WWTP and Biosolids/Sludge studies.

A third and ongoing sampling campaign, Land Application of Biosolids Containing PFAS Interim Strategy, was implemented in the summer of 2021. Dissimilar to the first two EGLE-coordinated sampling campaigns where all samples were collected by dedicated staff and analyzed by a single laboratory, the Land Application of Biosolids Containing PFAS Interim Strategy requires WWTP operators intending to land apply biosolids to independently sample the final treated solids generated at their facilities and send these samples to a PFAS-certified laboratory [15]. Once analyzed, laboratory reports are submitted to EGLE individually by WWTP operators. Results of sample analyses may limit or prohibit disposal method (e.g., land application) and/or lead to source investigations [15]. WWTP sampling frequency is contingent on

treatment volume, permit, if the WWTP has an Industrial Pretreatment Program (IPP), and previous interim strategy PFAS results [15]. At the time of procurement, the Land Application of Biosolids Containing PFAS Interim Strategy dataset contained nearly 300 samples from 181 WWTPs. Collectively, 406 solids samples were made available for analysis through a data sharing agreement between EGLE and the authors of this work.

2.2. Biosolids terminology

The term “biosolids” has varying definitions between researchers, WWTP operators, and state and federal regulatory agencies. USEPA establishes requirements for biosolids classes (i.e., A or B) and disposal methods under Title 40 of the Code of Federal Regulations (CFR) Part 503 [91]. The state of Michigan is 1 of 9 states delegated to implement the biosolids program as part of their National Pollutant and Discharge Elimination System (NPDES) program. Michigan’s Biosolids Program is regulated by EGLE’s Water Resources Division (WRD), under the authority of Part 31 of the Water Resources Protection of the Natural Resources and Environmental Protection Act, 1994 PA 451, Michigan’s Part 24 Administrative Rules ([92]). At the time of writing, the Land Application of Biosolids Containing PFAS Interim Strategy restricts the land application of biosolids if they are above an industrially impacted threshold for PFOS [15]. All final treated solids are referred to as biosolids in this study regardless of the disposal method (i.e., land application, landfilling, incineration) (Fig. S1).

2.3. Sample collection and laboratory analytical methods

EGLE, their contractors, and WWTP operators sampled biosolids in

Table 1

Compilation of literature studies investigating PFAS in biosolids (n = 30 studies) with select results relevant to the results of this study. A larger version of this table (which includes the full suite of analytes and number of samples) can be found in the [Supplementary Information \(Table S1\)](#). All concentrations are ng/g dry weight.

Study (Country)	No. of Analytes	Sampling Year (s)	Mean Σ PFAS	Σ PFAS Range	PFHxA	EtFOSAA	MeFOSAA	PFOS
[39] (U.S.)	13	2001	539	NA	3–12	NA	NA	308–618
[72] (Canada)	13	2002	NA	NA	NA	NA	NA	0.1–460
[62] (U.S.)*	12	1998–2004	617	176–3390	NA	ND-544	ND-154	14–2610
[73] (U.S.)	12	2004	NA	NA	NA	91–100	130–140	81–160
[74] (U.S.)	15	2004–2005	NA	NA	NA	NA	NA	26–65
[75] (U.S.)*	8	2005	166	17–382	NA	NA	NA	8.2–110
[28] (U.S.)	16	2004–2007	436	NA	NA	42–72	63–143	80–219
[76] (Singapore)	2	2006–2007	NA	NA	NA	NA	NA	31–702
[69] (U.S.)	12	2005–2013	NA	NA	ND-21	NA	NA	ND-68
[77] (Hong Kong)	19	2008	NA	NA	0.3–28	NA	NA	3–7304
[68] (China)*	16	2009	19	6.8–66.7	NA	NA	NA	0.9–22.5
[78] (Thailand)	10	2009	NA	NA	0.3–100	NA	NA	396–526
[79] (Spain)	5	-	NA	0.5–39	NA	NA	NA	ND-5
[71] (Korea)	15	2010	NA	NA	NA	NA	NA	15–260
[34] (Greece)	4	2009–2010	NA	NA	0–62	ND-11.2	ND-9.9	2–17
[80] (China)	14	2010	NA	126–809	0.5–100	NA	NA	28–173
[81] (China)	16	2010–2011	NA	NA	NA	NA	NA	ND-218
[70] (Germany)	11	2008–2013	NA	NA	ND-680	NA	NA	ND-7600
[82] (Nigeria)	10	2012	NA	NA	ND-0.25	NA	NA	ND-0.5
[83] (Switzerland)	10	2011	NA	NA	1–91	NA	NA	197–324
[84] (Kenya)	9	2013	NA	NA	ND-0.56	NA	NA	ND-0.67
[85] (Spain)	20	-	NA	NA	ND-32	NA	NA	ND-84
[8] (Australia)*	14	2014	128	0–403	ND-12	NA	NA	0–370
[86] (Sweden)*	79	2004–2017	337	49.7–1124	ND-7.2	ND-49	ND-21	6.2–41
[87] (Germany)	14	2010–2016	NA	NA	ND-16	NA	NA	ND-698
[27] (Australia)* *	160	1976–2017	307	13–900	0.6–3.2	1.9–582	2.2–41	0.35–284
[36] (Australia)	21	2017	34	2–130	ND-13	NA	NA	ND-90
[67] (Canada)*	24	2012–2017	30	3.4–53	0.2–4.7	NA	NA	0.49–50
[38] (Australia)*	44	2018	250	NA	ND-17	ND-50	ND-56	0.9–190
[88] (U.S.)* **	92	2021	78–584	NA	15	3.9	6	19

NA – not analyzed

ND – non-detect (includes values <MDL, <RL, and <LOQ).

*reported mean Σ PFAS and/or range

* *determined in this study using select samples classified as sewage sludge from [supporting materials](#).

* ** mean concentration used when range is unavailable

accordance with the state “Biosolids and Sludge PFAS Sampling Guidance” [93]. This sampling procedure involves the use of PFAS-free equipment, decontamination between samples, collection of field and equipment blanks, and preservation of collected samples at temperatures $\leq 6^\circ\text{C}$ prior to laboratory analysis. The “PFAS Minimum Laboratory Analyte List” developed by the state of Michigan outlines the recommended analytes for biosolids [89]. The minimum analyte list increased from 24 to 28 in 2019 [94]. PFECHS was included as an additional analyte (for a total of 29 PFAS) in the [42] Statewide WWTP and Biosolids/Sludge Study. Due to very low detection rates of the more recently analyzed PFAS, we only analyzed the initial 24 compounds included in all biosolids samples. Laboratories using 537 Modified, a modified version of the USEPA 537 Rev. 1.1 but not approved by the USEPA, and ASTM D-7968 with isotope dilution are recommended for PFAS analysis for solid matrices [93].

The use of 537 Modified and ASTM D-7968 for PFAS quantification in solid matrices involves solvent (e.g., methanol) extraction from the solid matrix followed by liquid chromatography-tandem mass spectrometry (LC/MS-MS) analysis. Sample preparation under 537 Modified uses solid-phase extraction prior to LC/MS-MS analysis, while ASTM D-7968 centrifuges each sample prior to filtering and adjusts pH using an acetic acid solution [95]. To meet the state recommended minimum analyte list, modified versions of these methods must include additional surrogate standards. Inconsistent analyte recovery due to matrix effects often leads laboratories to make corrections using isotopic dilution [96].

Approximately 55% of samples were analyzed using ASTM D-7968 (modified to include surrogates for the additional PFAS analytes recommended by EGLE), 42% using 537 Modified, and 3% either used other methods or did not list the analytical method in the laboratory report. Although variations between laboratories may exist, the greatest variability likely occurs during the methanol extraction process, as the participating laboratories employ proprietary procedures and do not disclose specifics. Approximately 82% of samples in this study were performed by laboratories utilizing isotope dilution to account for varying degrees of analyte recovery.

2.4. Study sample population and quality assurance/quality control

All PFAS analyses of solids samples in this study rely on internal quality control and quality assurance measures performed by each laboratory. Laboratory reports for several of the participating laboratories indicated the use of a method blank, laboratory control sample, and laboratory control sample duplicate. Data collected during the Evaluation of PFAS in Influent, Effluent, and Residuals of WWTPs in Michigan study were subjected to additional internal quality assurance and quality control (QA/QC) measures taken by EGLE and their contractors to ensure that compiled PFAS concentrations match the original laboratory reports [25]. A similar QA/QC process was performed in this study on samples from the 2021 Statewide WWTP and Biosolids/Sludge Study and Land Application of Biosolids Containing PFAS Interim Strategy by comparing the compiled concentrations of each sample to those of the original laboratory reports. Inter-laboratory comparisons were not performed for these data, as (1) both EGLE studies utilized only a single laboratory, and (2) WWTP operators participating in the Biosolids/Sludge Study and Land Application of Biosolids Containing PFAS Interim Strategy submitted samples independently to one of eight PFAS-certified laboratories. However, PFAS detection rates were evaluated between laboratories in our study as an additional examination of the data. One laboratory was found to exhibit substantially above average detection rates for all analytes relative to the other laboratories (Fig. S2), and this led to the disqualification of all 11 samples analyzed by this laboratory.

Samples included in this study are limited to solid samples representative of final treated solids analyzed by laboratories using either ASTM D7968 or 537 Modified (Table S3) with acceptable average detection rates (Fig. S2) and verifiable laboratory reports. Discarded

samples were comprised of incinerator ash, primary or secondary clarifier sludge, lacked laboratory reports or laboratory reports that did not specify the analytical method, or in some cases, an incorrect laboratory method for biosolids (e.g., ASTM D7979–17), or listed non-detect for all PFAS analytes. This process yielded a total study population of 350 solids samples for 190 WWTPs. The number of the biosolids samples per WWTP is illustrated in Fig. S3. Characteristics of each WWTPs, such as population equivalent, wastewater design capacity, known industries with regulated discharges, treatment processes, and rates of solids generation have neither been collected nor compiled at a state-level.

2.5. Statistical analyses

Standard statistical analyses were based on carbon-fluorine moiety (i.e., carbon chain length), functional group, total and individual PFAS constituent concentrations, and compound detection rates. Long and short-chain classification is consistent with previous studies [5], where long-chain compounds are $\geq\text{C}_6$ and $\geq\text{C}_8$ for PFAS with sulfonic and carboxylic functional groups, respectively. Conversely, short-chain PFAS compounds consist of $\leq\text{C}_5$ for sulfonic functional groups and $\leq\text{C}_7$ for carboxyl functional groups (Table S2). Where relevant, sample concentrations reported below the laboratory reporting limit were considered non-detect (ND) and converted to 0 ng/g concentrations. This method for evaluating censored data generally biases analytical results low [97]; however, we consider it appropriate as reporting limits were typically ~ 1 ng/g or less. Variances of 24 PFAS compounds in the sample population were analyzed using principal component analysis (PCA) generated and plotted through the Factoshiny package from the FactoMineR library in Rstudio using imputed data. A hierarchical cluster analysis of the 24 PFAS compounds was performed using Minitab.

3. Results and discussion

3.1. Trends in PFAS concentration

Summary statistics for all PFAS including detection rates and median, maximum, and average (with non-detects assigned 0 ng/g dry wt.) concentrations for all 350 biosolids samples from 190 WWTPs are provided in Table 2. Concentrations for all 24 PFAS above laboratory reporting limits are illustrated as box plots showing median (red line), interquartile range (blue box), 95% confidence intervals (dash black line), and outliers (red + symbol) (Fig. 1). The symmetry of median concentrations relative to interquartile ranges and confidence intervals in log space suggest the concentrations for all individual PFAS are lognormally distributed. These data were further assessed via Minitab generated probability plots for 21 PFAS in the dataset containing a sufficient number of detections for distributional analysis. Of these PFAS, 15 (71%) are best described by a lognormal distribution according to visual inspection and a modified Anderson-Darling (AD) metric ≤ 1.4 . Distributions of the other 6 PFAS (PFTeDA, PFUnDA, PFOS, N-EtFOSAA, 8:2 FTSA, 6:2 FTSA, and PFHxA) are closest to lognormal in distribution ($1.8 \leq \text{AD} \leq 3.4$), but exhibit greater deviations from the theoretical distribution caused by outlier concentrations (Fig. 1). Limited detection of PFNS (1.7%), PFPeS (0.003%) and 4:2 FTSA (0.003%) precluded assessments of distributional fit, and resulted in undefined interquartile and 95% confidence intervals for PFPeS and 4:2 FTSA (Fig. 1b).

Of the PFCAAs shown in Fig. 1a, PFTeDA (C14, 21% detection) has the lowest median concentration of 1.5 ng/g dry wt. and the smallest interquartile range and confidence interval. The commonly detected PFDA (C10, 73% detection) has the highest median concentration of 5.1 ng/g dry wt., greatest interquartile range, and largest confidence interval out of the PFCAAs. PFOS was the most ubiquitously detected PFAS and detected in 95% of samples (Fig. 1b). PFOS also has the highest median (11 ng/g dry wt.) and maximum (2150 ng/g dry wt.) concentrations of the entire dataset, and the largest confidence interval. Concentrations in at least one biosolids sample are greater than: (i)

Table 2

Compound specific statistics in sample population (n = 350) organized by PFAS class and sub-class (long- and short-chain designation). Non-detections are imputed as zeros. Class and sub-class rows describe collective statistics and are bolded. Concentrations are all on a dry-weight basis and mean values with $\pm$ standard deviation included.

Class	Homologue	Detection Rate (%)	Mean (ng/g dry wt.)	Median (ng/g dry wt.)	Maximum (ng/g dry wt.)
PFSA		96	43 $\pm$ 180	14	2158
Long-chain		95	42 $\pm$ 180	12	2158
	PFDS	36	1.8 $\pm$ 5.6	0	54
	PFNS	1.7	0.0 $\pm$ 0.3	0	4.7
	PFOS	95	40 $\pm$ 179	10	2150
	PFHpS	1.4	0.0 $\pm$ 0.4	0	7.4
	PFHxS	14	0.3 $\pm$ 1.2	0	11
Short-chain		21	1.2 $\pm$ 3.5	0	28
	PFPeS	0.3	0.0 $\pm$ 0.1	0	1.9
	PFBS	21	1.2 $\pm$ 3.5	0	28
PFCA		89	28 $\pm$ 51	11	389
Long-chain		79	18 $\pm$ 35	6.5	298
	PFTeDA	21	1.0 $\pm$ 11	0	200
	PFTTrDA	10	0.4 $\pm$ 2.5	0	35
	PFDODA	62	2.4 $\pm$ 4.8	1.3	55
	PFUnDA	37	1.4 $\pm$ 5.2	0	78
	PFDA	73	6.8 $\pm$ 13	2.7	120
	PFNA	36	1.2 $\pm$ 3.8	0	40
	PFOA	63	4.8 $\pm$ 11	1.2	96
Short-chain		77	10 $\pm$ 30	3.0	377
	PFHpA	14	0.8 $\pm$ 5.4	0	74
	PFHxA	76	5.1 $\pm$ 12	2.5	162
	PFPeA	39	3.4 $\pm$ 11	0	114
	PFBA	15	1.1 $\pm$ 4.1	0	38
Sulfonamide		94	25 $\pm$ 45	14	605
	MeFOSAA	92	13 $\pm$ 18	8.5	190
	EtFOSAA	87	10 $\pm$ 30	4.2	461
	FOSA	49	1.7 $\pm$ 4.6	0	73
Fluorotelomer		27	11 $\pm$ 122	0	2186
	8:2 FTSA	19	1.2 $\pm$ 8.0	0	136
	6:2 FTSA	17	9.8 $\pm$ 114	0	2050
	4:2 FTSA	0.3	0.0 $\pm$ 0.5	0	9.3

10 ng/g dry wt. for 20 of the 24 PFAS (PFNS, PFHpS, PFPeS and 4:2 FTSA being the exceptions and having the lowest detection), (ii) 100 ng/g dry wt. for 9 PFAS (PFTeTDA, PFDA, PFOS, N-MeFOSAA, N-EtFOSAA, 8:2 FTSA, 6:2 FTSA, PFHxA, PFPeA), and (iii) 1000 ng/g dry wt. for PFOS and 6:2 FTSA.

Note that 6:2 FTSA is a common replacement compound for legacy PFOS, indicating that the WWTPs with high 6:2 FTSA concentrations in biosolids have significant industrial wastewater contributions. These industrial trends are apparent in Fig. 2 which illustrates box plots of PFCA (n = 11), PFSA (n = 7), Sulfonamides (n = 3), and FTSA (n = 3) where compounds with sulfonic functional groups (PFSA, Sulfonamides, and FTSA) are prone to a greater frequency of high outlier concentrations. This is particularly noticeable for FTSA which has a median concentration nearly one order of magnitude less than PFCA, PFSA, and Sulfonamides, yet still has some of the highest concentration outliers. These concentration outliers are also responsible for PFSA and FTSA exhibiting more deviations from a lognormal distribution with AD values of greater than 3, whereas AD values for PFCA and Sulfonamides are 1.1 or less. The greatest interquartile range and widest confidence intervals in Fig. 2 are observed for PFCA presumably due to a larger number of detectable compounds (n = 11) relative to the other groups.

Similar to the individual PFAS, the distribution of total PFAS (Σ_{24} PFAS) from 190 Michigan WWTPs reasonably follows a lognormal distribution (AD=1.8) (Fig. 3a). Sample concentrations ranged between approximately 1 and 3200 ng/g dry weight (dry wt.) with all analytes (n = 24) detected at least once. The greatest contribution to Σ_{24} PFAS is from long-chain compounds which account for 88% of PFAS in biosolids on average (Table 3). The most frequently detected chain lengths include C8, C10, and C6 which contribute 70%, 9.6%, and 8.9% to Σ_{24} PFAS, respectively. Between the two functional groups, PFAS with sulfonates (e.g., PFSA, Sulfonamides, and FTSA) are more abundant, representing 71% of total PFAS concentration. Although landfill leachates are a significant source of PFAS entering some of the study WWTPs, the short-chain carboxyl chemistry associated with leachates (e.g., [8,9]) is distinct from the long-chain sulfonic composition observed in biosolids in this study and others (refer to Table 1).

We report lower average concentrations of Σ_{24} PFAS than other studies, yet the maximum concentration of 3192 ng/g dry wt. appears to be consistent (Table 1). Our mean concentration of 108 ± 277 ng/g dry wt. (Table 3) is considerably less than most of the Σ PFAS reported in other studies which are as high as 617 ng/g dry wt. (Table 1). Higher Σ PFAS concentrations for studies with larger analyte suites (e.g., [27,38,88]) are expected due to the inclusion of more compounds. To better understand this discrepancy, data representing biosolids from the supplementary data of Munoz et al. [27] and Thompson et al. [88] were extracted and evaluated. In the case of Munoz et al. [27], approximately 102 of the 160 analytes were detected at least once in the study's biosolids samples. Similar to what is found in Michigan (Table 2), the greatest contributions to Σ PFAS in those samples include PFOS and N-EtFOSAA; however, Σ PFAS contributions from PFOSAmS and 6:2 FTAB are 18% and 13%, respectively, which are not included in our study analyte suite. Likewise, Σ PFAS contributions of 36% and 12% were observed for 6:2 and 8:2 diPAP in samples from Thompson et al. [88]. Another reason for the lower mean Σ PFAS values in this study is the inclusion of samples from WWTPs of all sizes and industrial contributions. Unlike the majority of studies in Table 1 which focus on biosolids samples from industrially impacted WWTPs, samples in this study are collected from a diverse set of 190 WWTPs that include facilities serving communities as small as 700 residents with little or no industry, providing a more realistic representation of PFAS distributions and concentrations in biosolids across an entire U.S. state.

Compound-specific concentration ranges reported in Table 2 are also relatively consistent with other published values (Table 1 and S1). Armstrong et al. [69] suggested legacy 3 M compounds (i.e., PFOS) in biosolids have not substantially increased or declined since the early 2000s which is consistent with our study data. The PFOS concentration

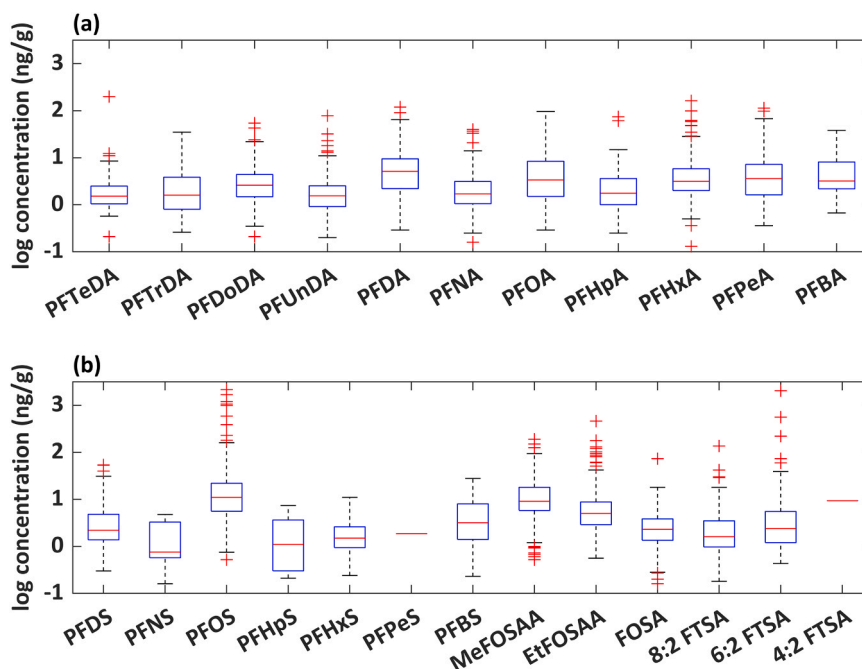


Fig. 1. Box plot of PFAS concentrations (ng/g) in biosolids denoting median (red line), quartile range (blue box), 95% confidence intervals (dashed line), and outlier (red +) for all (a) PFCAs and (b) PFSA, Sulfonamides, and FTSA detections. Undefined quartiles and confidence intervals for PFPeS and 4:2 FTSA are due to limited detection.

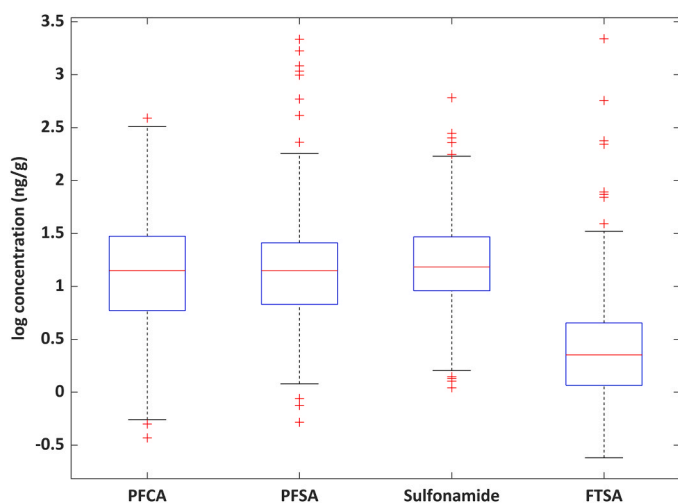


Fig. 2. Box plot of PFAS concentrations (ng/g) in biosolids denoting median (red line), quartile range (blue box), 95% confidence intervals (dashed line), and outlier (red +) for the 24 PFAS analytical suite grouped as PFCAs ($n = 11$), PFSA ($n = 7$), Sulfonamide ($n = 3$), and FTSA ($n = 3$).

range of ND–2150 ng/g dry wt., 40 ± 179 ng/g dry wt. mean concentration, and 95% detection rate indicates that it is consistently a dominant PFAS in 2018–2022 biosolids samples, consistent with many early studies [28,39,62] despite the voluntary manufacturing phase out of PFOS (and PFOA) by 3 M in the early 2000 s [5]. More recent studies report a narrower range in PFOS concentrations; however, many are sampled from Australian WWTPs which generally receive substantially less PFAS loading than the U.S. [27,36,38,98].

Munoz et al. [27] identified FTSA as widespread PFAS replacement compounds, but their findings are inconsistent with the data in this study as well as others [38,73]. After isolating the biosolids data from Munoz et al. (2021) (as it contains several organic waste products), we found Σ_{15} FTSA contribute just over 2% of total PFAS concentration.

We report a maximum concentration of 6:2 FTSA exceeding 2000 ng/g dry wt., but this was isolated to a single WWTP with heavy industrial inputs. Study data and those of past studies agree on the ubiquitous occurrence of sulfonamide compounds (i.e., N-MeFOSAA, N-EtFOSAA, and FOSA) in biosolids which are known PFOS precursors [27,28,38,62]. These trends are explored further in the following subsection.

Long and short-chain PFSA are most ubiquitously detected (96%) followed by sulfonamides (94%) and PFCAs (89%). This aligns well with many studies reporting 100% detection for several PFSA compounds – PFOS being most ubiquitously reported [28,34,39,62]. As a legacy PFAS, PFOS detections are expected to decrease over time, leading to diminished detection rates. As previously mentioned, the presence and abundance of PFOS in wastewater treatment has not substantially declined. We report 5% less frequent detections for this compound than previous studies which is most likely attributed to the diverse nature of the WWTPs included in this study, many of which service small communities and do not receive industrial wastewater or landfill leachate.

Of the highly detected compounds, the short-chain carboxyl chemistry of PFHxA is surprisingly detected in 76% of the samples and is roughly equivalent to the detection of long-chain carboxyls (79%), although the average concentration of long-chain carboxyls (18 ± 35 ng/g dry wt.) in biosolids is nearly four times greater than PFHxA. Some partitioning of PFHxA onto biosolids certainly occurs but higher detection and concentrations of PFHxA (5.1 ± 12 ng/g dry wt.) than PFOA (4.8 ± 11 ng/g dry wt.), a longer-chain compound with stronger partitioning behavior, is unexpected. Biosolids samples with higher PFHxA concentrations occur for WWTPs that receive larger volumes of landfill leachate, which tends to abound in PFHxA [2,8,9,99]. Other studies have observed similar trends [27,38,39,77,78] and many suggest the high frequency of detection for PFHxA may be attributed to precursor (e.g., fluorotelomer alcohols (FTOHs), FTSA, and diPAPs) transformations facilitated by microbial activity in activated sludge systems [64–66,100]. Expanding the PFAS analyte suite to include known PFHxA precursors would provide further insight into the importance of precursor transformation as a mechanism impacting the detection and relative abundance of PFHxA.

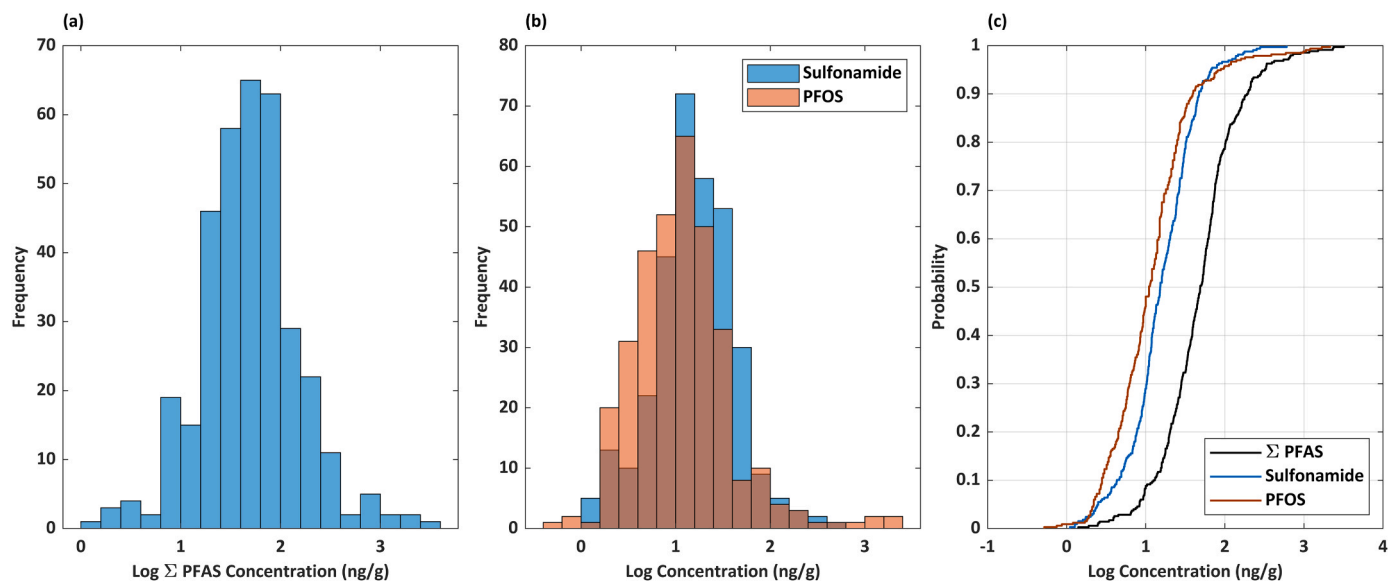


Fig. 3. (a) Histogram of total PFAS concentrations ($\Sigma 24$ PFAS) in sample population ($n = 350$), (b) histogram of Sulfonamide compounds (N-MeFOSAA, N-EtFOSAA, and FOSA) and PFOS for full sample population ($n = 350$), and (c) cumulative distribution function of all detected $\Sigma 24$ PFAS ($n = 350$), Sulfonamide ($n = 328$), and PFOS ($n = 333$) values.

Table 3

Summary statistics of total PFAS in sample population ($n = 350$). The % values refer to the percent of the samples in which each chain length and functional group were above detection. Mean concentrations include $\pm$ standard deviation.

Chain Length		Functional Group		Concentration Σ PFAS (ng/g dry wt.)				
Short	Long	Carboxyl	Sulfonic	Minimum	Mean	Median	Maximum	
12%	88%	29%	71%	1.4	108 ± 277	49	3192	
C4	C5	C6	C7	C8	C9	C10	C11	
2.3%	3%	8.9%	0.3%	70%	1%	9.6%	1.1%	
C12	C13	C14						
2.7%	0.2%	0.6%						

3.2. Prevalence of sulfonamide compounds

Collectively, the three sulfonamide compounds are detected as ubiquitously in biosolids as PFOS (Table 2). Of these compounds, N-MeFOSAA is most ubiquitously detected (92%), followed by N-EtFOSAA (87%), and FOSA (49%). Further investigation into the distribution of Σ_3 sulfonamide concentrations (FOSA+N-MeFOSAA+N-EtFOSAA) reveals greater contributions to Σ_{24} PFAS concentration than PFOS (Fig. 3b,c). Trends in median concentration, interquartile range, and confidence intervals are also consistent with the order of detection with N-MeFOSAA being the highest, followed by N-EtFOSAA, and FOSA as the lowest (Fig. 1b). Comparing the concentrations of sulfonamide compounds reported in early studies in the U.S. [28,62,73] to those found in this study, with samples collected from 2018 to 2022, shows that sulfonamides are just as prevalent, if not more, than in 1998–2007 (Table S1).

Transformation of sulfonamide PFAS to terminal PFAAs, particularly PFOS, has been observed to be microbially mediated in both activated sludge [101] and in soils [102]. Given this, and the concentrations of sulfonamide PFAS measured in the biosolids studied here (Fig. 2b), PFOS generation likely occurs anywhere in the environment these precursors are found. Thompson et al. [88] demonstrated that concentrations of PFAAs in biosolids sampled as they are generated at WWTPs are biased low, because precursors not analyzed will continue to breakdown over time and form PFAAs. As mentioned in Section 3.1, expanding the analyte suite to include intermediary precursors to PFOS (e.g., FOSAA and N-EtFOSE) would greatly improve the characterization of PFAS in biosolids and lead to more reliable estimates of PFAAs likely to be generated after disposal.

3.3. Trends in compound variance

Evaluation of the loading plot generated during principal component analysis (PCA) reveals several interesting trends regarding contributions of variance by each PFAS compound. Collectively, the first two dimensions, or principal components (PC1 and PC2, Fig. 4), describe approximately 35% of variance. The variation associated with PC1 is attributed largely to long-chain carboxyl compounds of chain lengths C8–C12 (note $\cos^2$ values in Fig. 4) which trend together along PC1 in quadrant 4. Almost perpendicular to this cluster ($\sim 88^\circ$) is a tighter cluster which trends along PC2 in quadrant 1. The compounds which make the greatest contributions to PC2 are short-chain carboxyl compounds (C4–C7) and PFOS. Although the physiochemical properties of PFOS differ from the other compounds in this group, it likely trends with this secondary variance due to the nearly ubiquitous detection of PFOS (95%). Otherwise, there is a clear distinction between the long and short-chain carboxyl chemistry of PC1 and PC2, respectively.

A scree plot illustrating the distribution of sample variance for each principal component indicates 80% of the cumulative variance in PFAS concentrations occurs between PC9 (78%) and PC10 (82%) (Fig. S4). Eigenvalues from PC1–PC9 were analyzed for interpretable trends (Fig. S5). Consistent with trends shown in the PCA loading plot (Fig. 4), PC1 indicates that the greatest sample variance (21%) is primarily controlled by longer-chain PFCAs (PFTrDA, PFDoDA, PFUnDA, PFDA, PFNA, PFOA) and to a lesser extent, the two most commonly detected sulfonamides (N-MeFOSSA, N-EtFOSSA). Secondary variance (14%) along PC2 is dominated by C4–C7 PFCAs, notably PFHpA, PFHxA, PFPeA, and PFBA. The next principal component, PC3, addresses tertiary controls on variance (9%) by long-chain PFSAs (PFDS, PFNS,

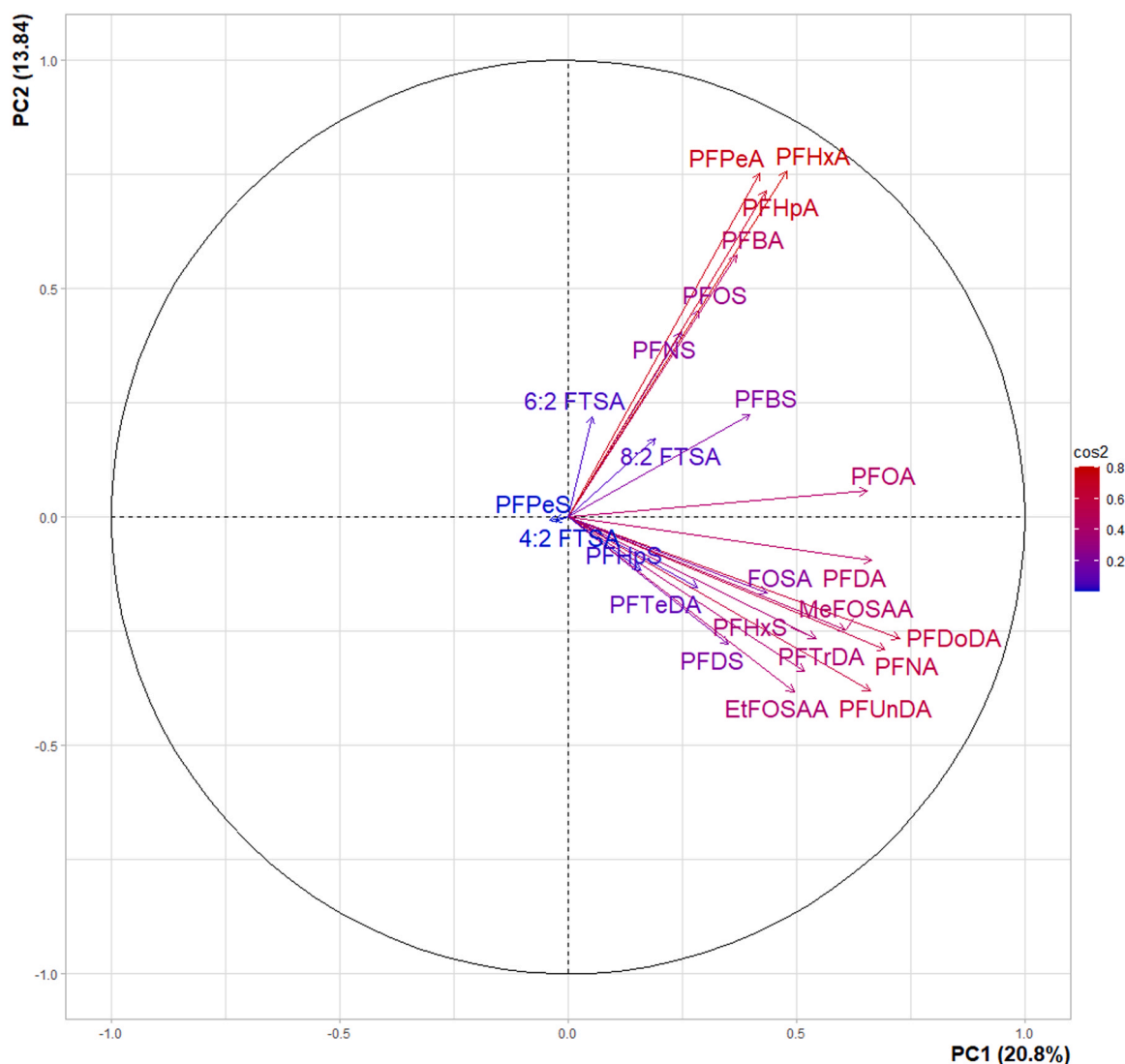


Fig. 4. Loading plot of constituents contributing to principal component analysis ($n = 350$). Vectors and $\cos^2$ values describe contributions to principal components 1 and 2 (PC1 and PC2, respectively). Note the dominance of long-chain compounds along PC1 and shorter-chain compounds along PC2.

PFOS) and FTSA (6:2 FTSA, 8:2 FTSA) given decreasing concentrations for the most commonly detected PFCAs (PFDoDA, PFDA, PFOA, PFHxA, PFPeA). Variance in PC4 (8%) encompasses trends of increasing concentrations of PFCAs with greater than 10 carbons (PFTeDA, PFTrDA, PFUnDA) in response to decreased concentrations of sulfonamides (FOSA, N-EtFOSAA, N-MeFOSAA) and FTSA (6:2 FTSA, 8:2 FTSA). Interestingly, PC5 appears to capture variance (7%) associated with increases in PFSA replacement compounds (6:2 FTSA, 8:2 FTSA, and PFBS) as concentrations of long-chain PFSA (PFDS, PFNS) and sulfonamides (FOSA, N-EtFOSAA, N-MeFOSAA) decrease. Trends in PCs 6 and 7 (6% and 5% variance) did not yield clear interpretations, and eigenvalues for PCs 8 and 9 (4% variance each) are highest for the two least detected PFAS in the dataset: 4:2 FTSA and PFPeS (Fig. S5).

Munoz et al. [27] performed PCA on PFAS in organic waste products (which included biosolids). Although their PC1 and PC2 describe more variance than this study ($\sim 47\%$), there are several important differences that may explain their higher variance along the first two principal components and that make it impossible to directly compare the results of the two studies. Most importantly, Munoz et al. [27] studied organic waste products which included sewage sludges, sludges mixed with compost of green waste, manures, and industrial sludges and ashes. Unlike this study, their PCA effectively describes variance of PFAS between multiple solid matrices – not exclusively biosolids (i.e., final

treated solid residuals from WWTPs). Additionally, unlike this study, the waste products in Munoz et al. [27] had been sampled over a period of several decades, which also imposes a temporal variability on the PCA captured influenced by historical usage of PFAS. Even in our study where samples were collected over a relatively narrow timeframe, variance in PFAS concentrations attributed to replacement compounds were observed. These factors, along with the limited sample size ($n = 47$) in Munoz et al. [27], means the variance captured in their PCA represents a substantially different statistical implication.

To further investigate trends in our study data, a cluster variable analysis was performed in Minitab comprised of 5 pre-set clusters formed using absolute distance correlation with complete linkage. A dendrogram of the clusters mostly supports the PCA interpretations (Fig. 5). The largest cluster (blue) on the left-hand side of the figure contains mostly long-chain PFCAs with clades (horizontal line grouping each leaf) linking PFTeDA-PFTrDA, PFDoDA-PFDA, PFUnDA-PFNA within a similarity range of 85–92%. Long-chain PFHxS and sulfonamide compounds are also located in the blue cluster, but exhibit less similarity. The MeFOSAA-EtFOSAA clade is located at less than 80% similarity, and FOSA and PFHxS remain as an isolated leaf until 64% and 77% similarity, respectively.

The second largest cluster (green), located on the right-hand side of Fig. 5, consists of PFCAs with chain lengths of C4–C8, with the exception

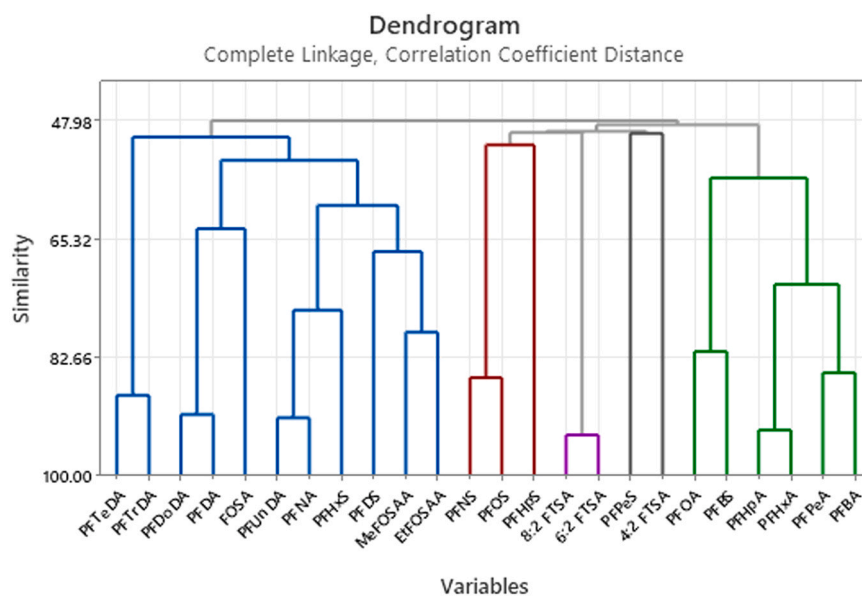


Fig. 5. Dendrogram using absolute distance correlation with complete linkage showing 5 clusters comprised of C9-C14 PFCAs, sulfonamides, and two PFSA (blue), C4-C8 PFCAs and PFBS (green), C7-C9 PFSA (red), FTSA (purple), and rarely detected PFPeS and 4:2 FTSA (black).

of PFBS. As expected, higher degrees of similarity are observed for PFHxA-FHhPa and PFBA-PFPeA than the PFOA-PFBS clade indicating 80% similarity. The third cluster (brown) is comprised of long-chain sulfonates PFNS, PFOS and PFHpS, with PFOS-PFNS paired at 84% similarity. The low detection rate 1.4% is responsible for PFHpS remaining as an isolated leaf with low (less than 50%) similarity with PFOS and PFNS. The fourth cluster (purple) consists of a 6:2 FTS-8:2 FTSA pair with very high similarity. The grouping and high similarity of these two FTSA is reasonable given that these compounds are often both present in formulations used to replace PFOS in various industrial applications and later generation AFFF. The fifth cluster (black) is comprised of PFPeS and 4:2 FTSA which were detected the least and remain as an isolated leaf until very low (50%) similarity.

3.4. PFAS enrichment in biosolids

The degree to which a PFAS compound favors the solid phase over the liquid phase determines the degree to which it will be enriched in biosolids. To evaluate the partitioning behavior of PFAS between wastewater influent and biosolids, samples of aqueous influent were paired with samples of biosolids from the same facility and collected on the same date ($n = 73$ pairs). Enrichment factors for individual PFAS are defined in this study as a dimensionless log ratio of biosolids concentration (converted from ng/g dry wt., parts-per-billion (ppb) to ng/kg dry wt., parts-per-trillion (ppt)) to influent aqueous concentration (ng/l, ppt). For example, a biosolids PFAS concentration of 50 ng/g dry wt. (5×10^4 ng/kg dry wt.) paired with an aqueous influent concentration of

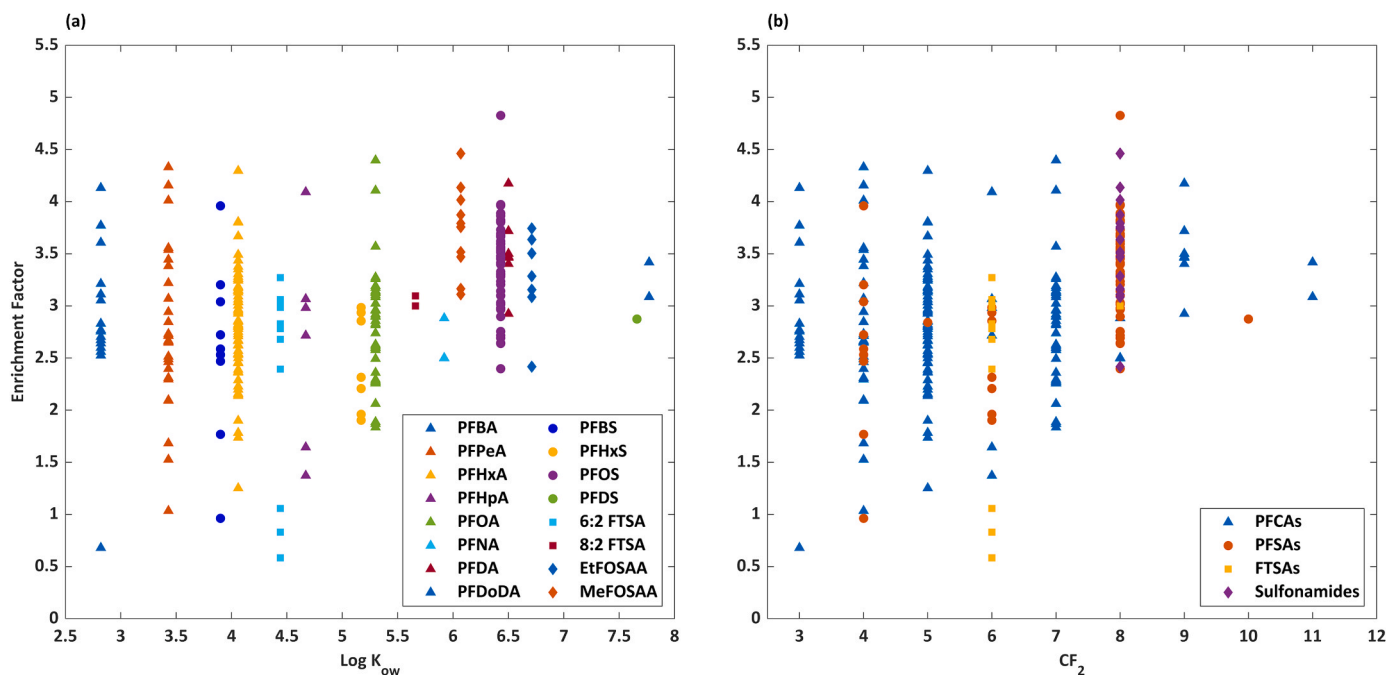


Fig. 6. Ratio of biosolids concentration (ng/kg) to influent concentration (ng/l) on a logarithmic scale describing enrichment of PFAS in biosolids versus (a) experimental $\log K_{ow}$ values and (b) carbon-fluorine units (CF_2).

5 ng/l yields a log enrichment factor of 4.

Enrichment factor is plotted against two proxies for hydrophobicity in Fig. 6. In Fig. 6a, log ratio of biosolids/influent concentration for 16 PFAS compounds (y-axis) is plotted against log values of experimental octanol-water partition coefficients (K_{ow} , x-axis) reported in the literature [103] (Table S4). Of the 73 data pairs used in the analysis, only five compounds in Fig. 6a contain at least 10 data pairs to delineate trends in PFAS partitioning: PFBA, PFPeA, PFHxA, PFOA, and PFOS. Enrichment factors for these compounds are distributed over an approximate range of 2.8–3.5 orders of magnitude. The lowest (minimum) ratio values for each compound/class exhibit the expected trend of longer-chain, more hydrophobic PFAS compounds partitioning to a greater extent to biosolids. The central tendency of the ranges, although not particularly well-defined by the data, also tend to mostly increase with increasing hydrophobicity as a function of chain length for the carboxyl compounds in order from PFBA (C4), PFPeA (C5), PFHxA (C6), PFOA (C8), and potentially PFDA (C10) although the central tendency for this compound is poorly resolved. Minimum and central tendency of PFOS are higher than PFOA, and N-MeFOSAA (limited data) tends to partition to biosolids more strongly than PFOS. All of these trends are consistent with recent studies indicating increased solids partitioning with increasing carbon chain length, and PFAS with sulfonic functional groups more strongly partitioning to solids than carboxylic functional groups [104,105]. Fig. 6b plots the enrichment factor for PFCA, PFSA, FTSA, and Sulfonamides against a second proxy for hydrophobicity, the number of carbon-fluorine units (i.e., carbons with ≥ 2 fluorine bonds), which is defined by the structure of each compound and not

experimentally derived. Despite the simplification of the PFAS data and different hydrophobicity proxy, trends in Fig. 6b mostly reflect those discussed in Fig. 6a with respect to minimum and central tendencies for PFCA and PFSA that generally indicate increased partitioning with CF_2 group/hydrophobicity. As also shown in Fig. 6a, trends in Fig. 6b are insufficiently resolved for FTSA and sulfonamides due to lack of data.

Previous studies have also reported broad ranges of PFAS partitioning behavior in WWTPs, deviating widely from their organic carbon-water partitioning coefficient (K_{oc}) values as sorption to sludge is not only driven by hydrophobicity, but also electrostatic interactions with amide groups and sorbed cations [63,106]. Additionally, differences in wastewater chemistry (pH, TDS, DOC, etc.), treatment processes, and solids handling and treatment all appear to influence partitioning making it difficult to understand the impact of any one parameter [63]. Differences in timescales between hydraulic and solids retention times in WWTPs also likely introduce uncertainty into the biosolids enrichment factors, despite the influent-biosolids pairs in our study having the same sampling date. For example, Vitale et al. [31] reported a solids retention time (11.2 d) for the Kalamazoo Water Reclamation Plant (an activated sludge plant, one of the 190 WWTPs contributing to biosolids samples in this study) that exceeds the plant hydraulic residence time (1 d) by an approximate factor of 10. This implies processes governing the partitioning of PFAS to biosolids operate over longer timescales than influent wastewater concentrations which are known to vary at least diurnally in WWTPs [107,108], and thus, PFAS concentrations in biosolids should exhibit less temporal variability than influent. This is shown by Vitale et al. [31] where coefficients of variation in background

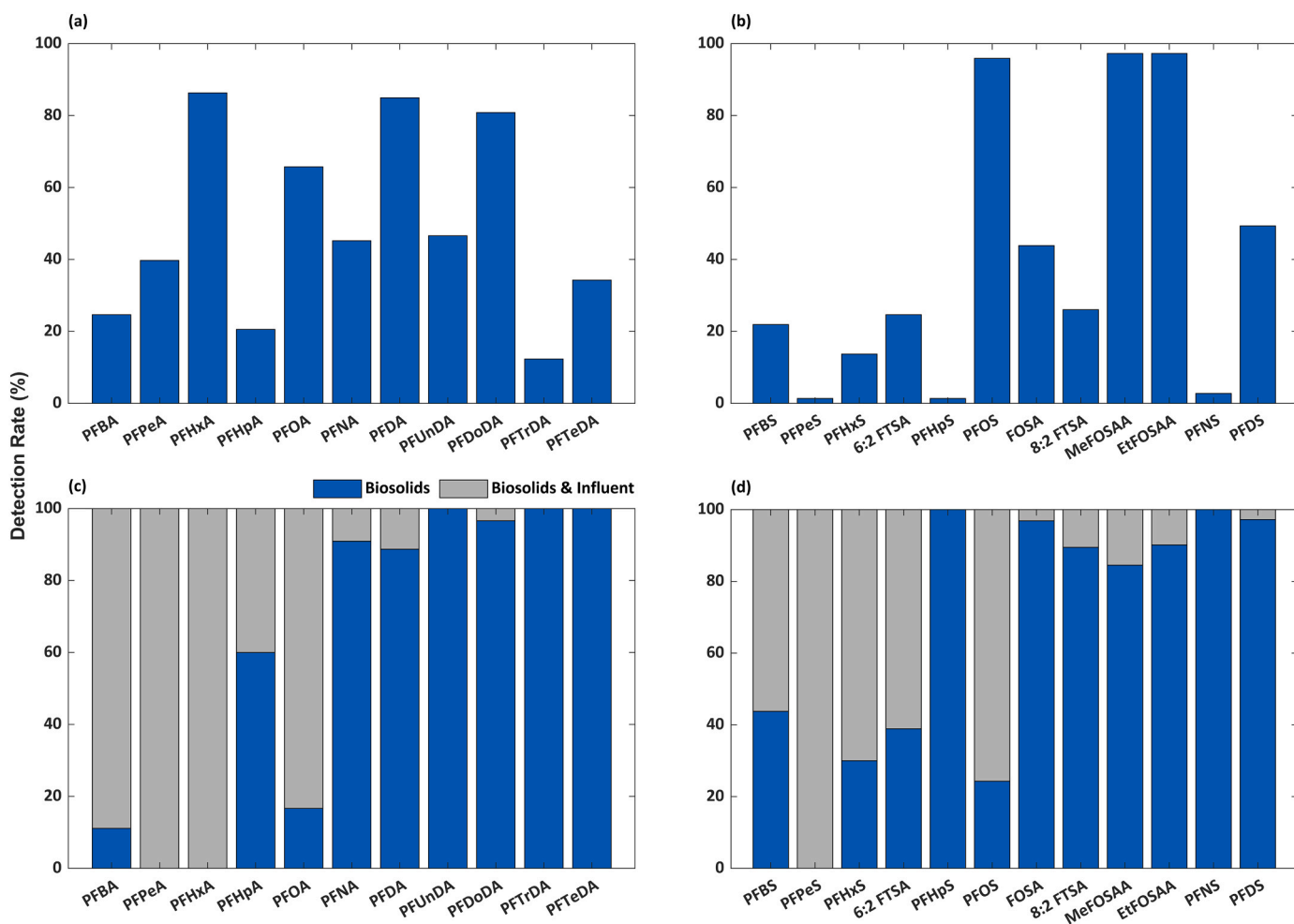


Fig. 7. Detection rates of individual PFAS from select samples ($n = 73$ pairs) categorized by (a) carboxylate and (b) sulfonic compounds. Plots (c) and (d) depict the frequency at which detections in (a) and (b) occur only in biosolids (blue) for both biosolids and wastewater influent (grey).

biosolids concentrations were nearly 4 times less than the influent.

Several compounds in the 73 aqueous influent-biosolids pairs have limited representation in Fig. 6, due to non-detection in either aqueous or solid matrices. This was observed most frequently for the less common, longer-chain compounds with very low solubilities, which made non-detection in the aqueous phase more likely. Biosolids detection rates of the 73 paired samples are illustrated in Fig. 7a for 11 carboxylate compounds and in Fig. 7b for 12 compounds with sulfonic functional groups. Due to the limited sample size, detections in Fig. 7a somewhat differ from the larger 350 sample population listed in Table 3. These data are further distinguished in Fig. 7c,d where PFAS detections are grouped as (i) limited to biosolids only (blue) or (ii) detected in both aqueous influent and biosolids (grey). Many PFAS, including PFBA, PFPeA, PFHxA, PFOA, PFPeS, and PFOS, were detected in both biosolids and aqueous influent for at least 75% of the samples. More specifically, PFHxA was detected in greater than 80% of the biosolids samples (Fig. 7a), and was always present in the aqueous influent when detected in biosolids (Fig. 7c).

Longer-chain compounds, such as PFUnDA, PFDODA, PFTrDA, PFTeDA, PFHpS, FOSA, PFNS, and PFDS, were below detection limits in 95% or more of the wastewater influent samples when detected in biosolids. For example, PFUnDA was detected in slightly over 40% of biosolids samples (Fig. 7a) but was never detected in the influent (Fig. 7c). While it is possible that some of the non-aqueous detections of PFUnDA and other long-chain compounds could at least partially be explained by transformations of undetected precursors during the wastewater treatment process, strong solids partitioning likely leads to these compounds being concentrated above the detection limit in biosolids. Maximum concentrations observed in biosolids for PFUnDA, PFTrDA, PFTeDA, PFHpS, and PFNS ranged from 5 to 200 ng/g dry wt. In the absence of influent detections for these compounds and assuming a conservative reporting limit of 1 ppt for influent, we can generally state these compounds are concentrated from a factor of 5 up to 5 orders-of-magnitude. Realistically, aqueous concentrations of these compounds are likely to be considerably less than 1 ppt with even greater enrichment in biosolids than listed above. Perhaps more importantly, these findings suggest that biosolids can serve as a useful indicator medium to test for the presence of these longer-chain, lower-solubility PFAS in influent waste streams.

3.5. Analyte limitations

To our knowledge, this study compiles the largest biosolids sample population analyzed for PFAS in the United States, and on a global scale is second only to Ulrich et al. [70]. This sample size captures the diversity of WWTPs throughout the state of Michigan (i.e., industrial impacts and lack thereof, absence/presence of contributions from landfill leachate, and community size), and is also representative of PFAS signatures in modern (2018–2022) biosolids. PFAS detection and quantification is limited to the analyte suite recommended by Michigan EGLE and lacks several important PFAA precursors shown in other studies to be frequently detected at appreciable concentrations.

The minimum PFAS analyte list increased from 24 to 28 analytes after the initial sampling campaign and is reflected in samples collected after 2019 [94]. Newly added are HFPO-DA (GenX), ADONA, 9Cl-PF3ONS, and 11Cl-PF3OUds. PFECs was also included in the [42] Statewide WWTP and Biosolids/Sludge Study, although it was detected only twice in samples taken from the same WWTP. Between the two samples, the mean concentration of PFECs was only 1.17 ± 0.15 ng/g and suggests the exclusion of this compound has a negligible impact on the scope of this work. HFPO-DA, 11Cl-PF3OUds, 9Cl-PF3ONS, and ADONA ($n = 336$, 96% of the samples) may be considered equally negligible, detected in 3.6%, 5.4%, 4.8% and 3.3% of the samples, respectively, at concentrations only slightly above the detection limit.

Perhaps less negligible are polyfluoroalkyl phosphate diesters (diPAPs), fluorotelomer alcohols (FTOHs), and fluorotelomer carboxylic

acids (FTCAs). Although we are unable to determine their relevance in Michigan WWTPs, Yoo et al. [61] detected FTOHs in biosolids-amended soils, suggesting their prevalence in the applied biosolids. Moodie et al. [38] reported the presence of FTCAs and diPAPs in biosolids with each class having over 90% detections and mean concentrations of 24 and 140 ng/g, respectively. Similar findings are reported by D'eon et al., [72], Fredriksson et al. [86], and Thompson et al. [88].

Although a 24-analyte suite is smaller than that used by Munoz et al. [27], Moodie et al., [38], and Thompson et al. [88], this study contains a larger analyte suite than roughly 87% of the biosolids studies reviewed (Table 1). Furthermore, many regulatory analytical suites for PFAS, including the upcoming USEPA Method 1633 [109], do not include many of the compounds missing from our suite (e.g., diPAPs and FTOHs). While this does not justify the absence of relevant PFAS compounds in this study, it brings to light the challenges facing this area of research going forward. We identify several instances where a more comprehensive analytical suite that includes FTOHs, FTCAs, and diPAPs would improve our characterization and understanding of PFAS in biosolids. Not only do these PFAS represent PFAA precursors, but quantifying their presence in WWTP matrices can improve mass balance estimates [31]. Currently, studies must attribute discrepancies between influent and effluent PFAS mass to error. Although it can be inferred that an increase in PFAA mass is due to transformations within WWTPs, inclusion of the aforementioned precursors would serve as quantitative evidence for this and aid in constraining error.

4. Conclusions

The presence of PFAS in WWTP influent, effluent and solid residuals/biosolids has been well established for at least 2 decades, yet there is still considerable progress to be made. In this study, we developed a comprehensive baseline for the occurrence of PFAS in biosolids, determined how these compounds vary throughout a geographically significant region (i.e., the state of Michigan), and address the need for additional analytes to better understand the implications associated with our results. The following are the key findings from this study:

- Biosolids studies relying on multiple data sources to form large sample populations require careful consideration of the contributing laboratories and the analytical methods to ensure comparability and statistical integrity of the data. Furthermore, it is crucial to exclusively focus on samples which represent fully processed solids (i.e., biosolids) at the end of the wastewater treatment to best represent PFAS concentrations prior to disposal.
- Consistent with previous studies, biosolids from Michigan WWTPs are dominated by long-chain sulfonic PFAS compounds, although select short-chain carboxyl compounds (PFHxA and PFPeA to a lesser extent) are frequently detected and can have concentrations similar to PFOA. Terminal short-chain carboxyl compounds may represent the occurrence of transformations in activated sludges and should receive additional attention in future investigations.
- The usage of PFSA replacements compounds (6:2 FTSA, 8:2 FTSA, and PFBS) to long-chain PFSA (PFDS, PFNS) and sulfonamides (FOSA, N-EtFOSAA, N-MeFOSAA) was also observed to contribute to variance in biosolids concentrations.
- PFOS remains a leading PFAS constituent in 2018–2022 biosolids throughout Michigan, and was detected in 95% of the samples despite the voluntary phase-out in manufacturing in the early 2000s.
- Sulfonamide compounds substantially contribute to Σ_{24} PFAS and are as ubiquitously detected as PFOS. Considering the implications of sulfonamide transformations to PFOS after disposal, additional research and regulatory focus on these compounds is needed.
- State-wide variability of PFAS compounds in biosolids is primarily controlled by carboxyl compounds of chain lengths C8–C12, and secondarily controlled by short-chain carboxyls and PFOS.

- Our data indicate trends in PFAS enrichment in biosolids for individual PFAS vary at least 3 orders-of-magnitude and generally increase with compound hydrophobicity; however, the partitioning of PFAS onto solids in WWTPs is a complex process not easily described nor constrained using experimentally-derived partitioning coefficients (i.e., K_{ow} and K_d).

This study focuses exclusively on the characterization of PFAS in fresh biosolids generated at the end of the wastewater treatment process. Substantial future work is necessary to fully develop an understanding of PFAS cycling and occurrence in WWTPs, biosolids, and the environment, particularly with respect to PFAS bioavailability, transformation, fate, and transport after land application. We encourage researchers, regulators, and others to improve upon these findings through the addition of relevant precursors, such as diPAPs, FTCAs, and FTOHs, and to sample final treated solids as frequently as financially possible.

Environmental implication

Biosolids generated from wastewater treatment generally become enriched in PFAS, an emerging environmentally-persistent contaminant known to cause adverse health impacts to humans and bioaccumulate within ecosystems. Biosolids are then either land applied or landfilled, and in some cases, incinerated, providing pathways for which PFAS can continuously cycle between engineered systems and the environment. This study investigates the composition, distribution, and variability of PFAS in the the largest known biosolids sample population collected in the U.S. to date, representing a geographically-diverse range of WWTPs serving communities of various sizes and industrial activity.

CRediT authorship contribution statement

Garrett W. Link: Conceptualization, Data Curation, Formal Analysis, Methodology, Validation, Visualization, Writing-Original draft; **Donald M. Reeves:** Conceptualization, Formal Analysis, Methodology, Visualization, Writing-Review and Editing; **Daniel P. Cassidy:** Conceptualization, Writing-Review and Editing; **Ethan S. Coffin:** Conceptualization, Writing-Review and Editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Data archive

https://github.com/dmreeves1/JHM_PFAS_Biosolids.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jhazmat.2023.132734](https://doi.org/10.1016/j.jhazmat.2023.132734).

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